

# Two Methods for Running Jupyter Notebooks.

In this document two methods for running Jupyter Notebooks are presented. The first method involves downloading and installing Miniconda. The notebook then runs locally on your machine. The second method involves Google Colab. In this method no software is downloaded or installed on your machine, and the notebook is hosted in the cloud.

## Instructions for installing Miniconda and Running a Jupyter Notebook on a Windows machine.

Miniconda is a lightweight version of Anaconda which is a distribution of Python. Jupyter notebooks provide an interactive platform for running python code.

In this document:

- Steps to download and install Miniconda
- Setting up a virtual environment
- Steps to download the packages required for ENEL 441 and ENEL ???
- Steps to run a Jupyter Notebook

## Downloading and Installing Miniconda

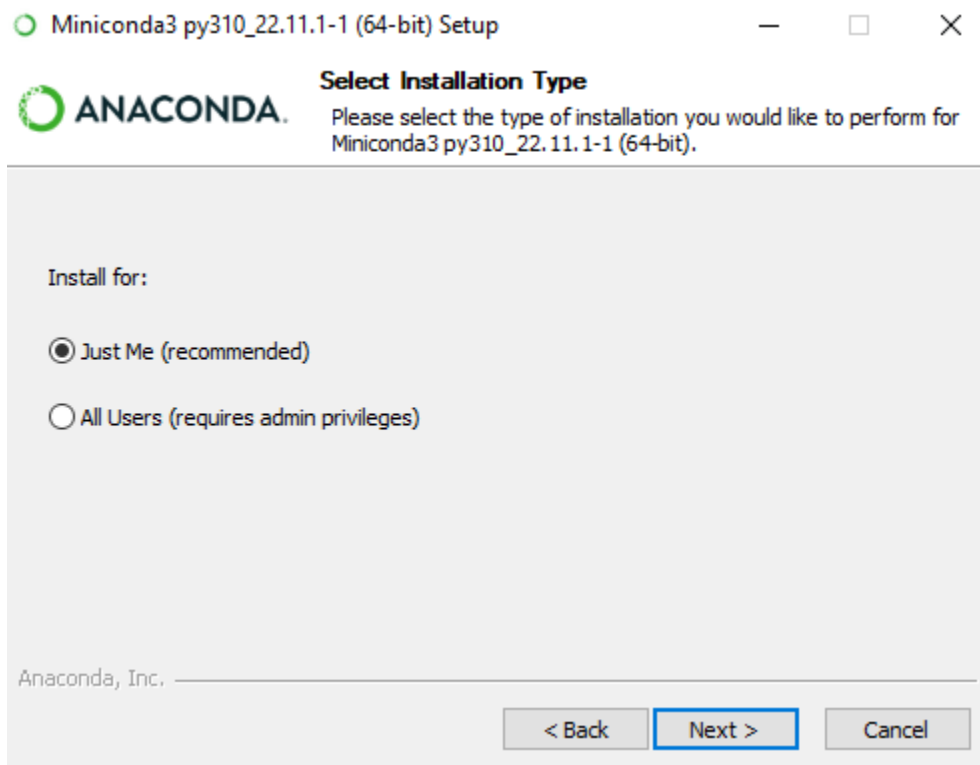
1. Go to the website: <https://docs.conda.io/en/latest/miniconda.html> and click on link to download windows 64bit version of the installer:

### Latest Miniconda Installer Links

*Latest - Conda 22.11.1 Python 3.10.8 released December 22, 2022*

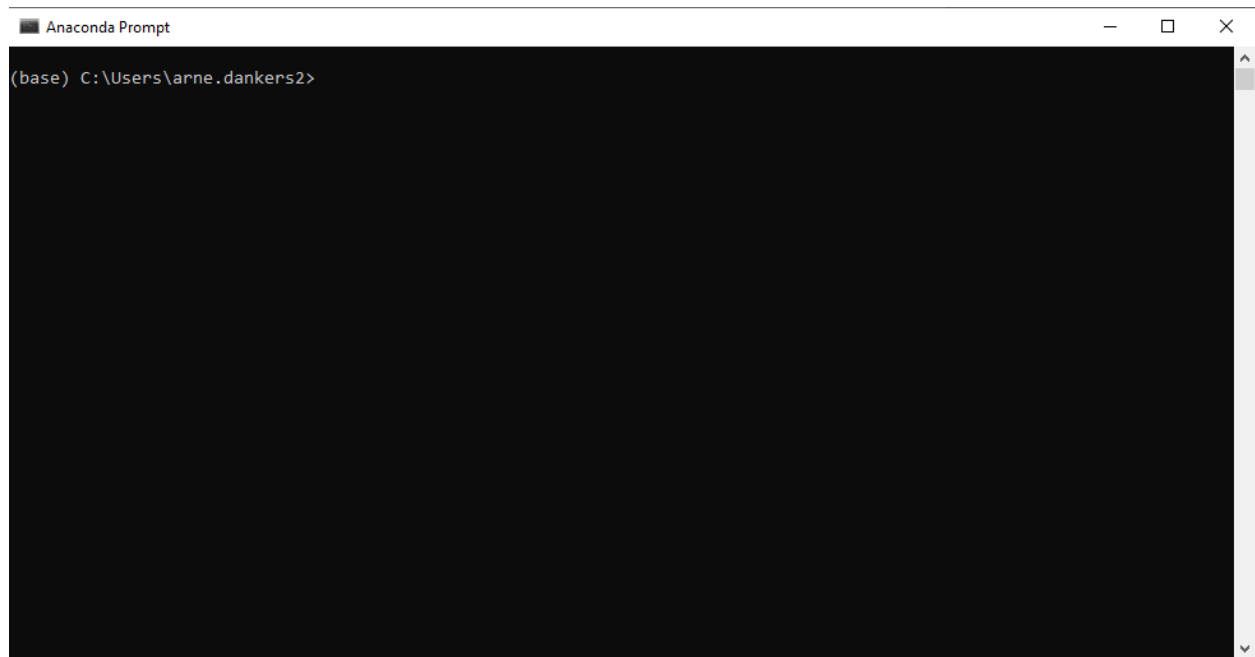
| Platform | Name                                   | SHA256 hash                                                      |
|----------|----------------------------------------|------------------------------------------------------------------|
| Windows  | Miniconda3 Windows 64-bit              | 2e3086630fa3fae7636432a954be530c88d0705fce497120d56e0f5d865b0d51 |
|          | Miniconda3 Windows 32-bit              | 4fb64e6c9c28b88beab16994bfba4829110ea3145baa60bda5344174ab65d462 |
| macOS    | Miniconda3 macOS Intel x86 64-bit bash | 7406579393427eaf9bc0e094dcd3c66d1e1b93ee9db4e7686d0a72ea5d7c0ce5 |

Run the installer. Make sure you run it locally (this should be the default).



Finish installation using all the defaults.

Click on the Windows Start Menu, open Anaconda Prompt. This should open a window that looks like this:



When using Python it is a good idea to create Virtual Environments. This allows you to download different versions for packages in each virtual environment. In the next steps we will create a new virtual environment where we will download all the packages for the ENEL441 course. To do this: at the prompt type in:

```
>>conda create -n enel441 python=3.9
```

You will be prompted to proceed, type 'y' for yes.

This creates a virtual environment called enel441.

```
(base) C:\Users\arne.dankers2>conda create -n enel441 python=3.9
Collecting package metadata (current_repodata.json): done
Solving environment: done

## Package Plan ##

  environment location: C:\Users\arne.dankers2\AppData\Local\miniconda3\envs\enel441

added / updated specs:
- python=3.9

The following packages will be downloaded:

package | build | size
-----|-----|-----
certifi-2022.12.7 | py39haa95532_0 | 149 KB
pip-22.3.1 | py39haa95532_0 | 2.7 MB
python-3.9.15 | h6244533_2 | 19.4 MB
setuptools-65.5.0 | py39haa95532_0 | 1.1 MB
wincertstore-0.2 | py39haa95532_2 | 15 KB
-----|-----|-----
Total: | | 23.5 MB

The following NEW packages will be INSTALLED:

ca-certificates | pkgs/main/win-64::ca-certificates-2022.10.11-haa95532_0
certifi | pkgs/main/win-64::certifi-2022.12.7-py39haa95532_0
openssl | pkgs/main/win-64::openssl-1.1.1s-h2bbff1b_0
pip | pkgs/main/win-64::pip-22.3.1-py39haa95532_0
python | pkgs/main/win-64::python-3.9.15-h6244533_2
setuptools | pkgs/main/win-64::setuptools-65.5.0-py39haa95532_0
sqlite | pkgs/main/win-64::sqlite-3.40.0-h2bbff1b_0
tzdata | pkgs/main/noarch::tzdata-2022g-h04d1e81_0
vc | pkgs/main/win-64::vc-14.2-h21ff451_1
vs2015_runtime | pkgs/main/win-64::vs2015_runtime-14.27.29016-h5e58377_2
wheel | pkgs/main/noarch::wheel-0.37.1-pyhd3eb1b0_0
wincertstore | pkgs/main/win-64::wincertstore-0.2-py39haa95532_2

Proceed ([y]/n)? y

Downloading and Extracting Packages
python-3.9.15 | 19.4 MB | #####3 | 70%
pip-22.3.1 | 2.7 MB | #####9 | 85%
setuptools-65.5.0 | 1.1 MB | ##### | 100%
certifi-2022.12.7 | 149 KB | ##### | 100%
wincertstore-0.2 | 15 KB | ##### | 100%
```

Now we will activate the environment. Type:

```
>>conda activate enel441
```

```
(base) C:\Users\arne.dankers2>conda activate enel441
(enel441) C:\Users\arne.dankers2>
```

Note that the text in bracket preceding the prompt has changed. This indicates which environment is active.

Now we will install the required packages.

```
>>conda install numpy jupyter scipy matplotlib
```

You will be asked to confirm the installation. Type 'y' in the prompt.

```
(enel441) C:\Users\arne.dankers2>conda install numpy jupyter matplotlib scipy
Collecting package metadata (current_repodata.json): done
Solving environment: done

## Package Plan ##

  environment location: C:\Users\arne.dankers2\AppData\Local\miniconda3\envs\enel441

added / updated specs:
- jupyter
- matplotlib
- numpy
- scipy

The following packages will be downloaded:
```

| package                     | build          |        |
|-----------------------------|----------------|--------|
| anyio-3.5.0                 | py39haa95532_0 | 167 KB |
| argon2-cffi-21.3.0          | pyhd3eb1b0_0   | 15 KB  |
| argon2-cffi-bindings-21.2.0 | py39h2bbff1b_0 | 36 KB  |
| asttokens-2.0.5             | pyhd3eb1b0_0   | 20 KB  |
| attrs-22.1.0                | py39haa95532_0 | 84 KB  |

Plus more text will appear below this.

Wait until you see the following text in the window:

```
Preparing transaction: done
Verifying transaction: done
Executing transaction: done

(enel441) C:\Users\arne.dankers2>
```

Now install control package:

```
>>conda install -c conda-forge control
```

```
(enel441) C:\Users\arne.dankers2>conda install -c conda-forge control
Collecting package metadata (current_repodata.json): done
Solving environment: done

## Package Plan ##

  environment location: C:\Users\arne.dankers2\AppData\Local\miniconda3\envs\enel441

added / updated specs:
- control

The following packages will be downloaded:
```

| package                                    | build             |         |             |  |
|--------------------------------------------|-------------------|---------|-------------|--|
| ca-certificates-2022.12.7                  | h5b45459_0        | 143 KB  | conda-forge |  |
| certifi-2022.12.7                          | pyhd8ed1ab_0      | 147 KB  | conda-forge |  |
| control-0.9.3.post2                        | pyhd8ed1ab_0      | 294 KB  | conda-forge |  |
| libblas-3.9.0                              | 1_h8933c1f_netlib | 193 KB  | conda-forge |  |
| libcbblas-3.9.0                            | 5_hd5c7e75_netlib | 95 KB   | conda-forge |  |
| libflang-11.0.1                            | h0e60522_20210131 | 1.1 MB  | conda-forge |  |
| liblapack-3.9.0                            | 5_hd5c7e75_netlib | 2.7 MB  | conda-forge |  |
| llvm-openmp-11.0.1                         | h2d74725_0        | 563 KB  | conda-forge |  |
| m2w64-gcc-libgfortran-5.3.0                | 6                 | 342 KB  | conda-forge |  |
| m2w64-gcc-libs-5.3.0                       | 7                 | 520 KB  | conda-forge |  |
| m2w64-gcc-libs-core-5.3.0                  | 7                 | 214 KB  | conda-forge |  |
| m2w64-gmp-6.1.0                            | 2                 | 726 KB  | conda-forge |  |
| m2w64-libwinpthread-git-5.0.0.4634.697f757 | 2                 | 31 KB   | conda-forge |  |
| msys2-conda-epoch-20160418                 | 1                 | 3 KB    | conda-forge |  |
| numpy-1.24.1                               | py39hbccbffa_0    | 5.6 MB  | conda-forge |  |
| openssl-1.1.1s                             | hcfcb64_1         | 5.1 MB  | conda-forge |  |
| python_abi-3.9                             | 2_cp39            | 4 KB    | conda-forge |  |
| slycot-0.5.3                               | py39h73a3b6d_0    | 1.5 MB  | conda-forge |  |
| ucrt-10.0.22621.0                          | h57928b3_0        | 1.2 MB  | conda-forge |  |
| vs2015_runtime-14.32.31332                 | h1d6e394_9        | 1.2 MB  | conda-forge |  |
| Total:                                     |                   | 21.7 MB |             |  |

It will print a message about upgrading/downgrading some packages. Just type 'y' to accept.

```

The following NEW packages will be INSTALLED:

control                conda-forge/noarch::control-0.9.3.post2-pyhd8ed1ab_0
libblas                conda-forge/win-64::libblas-3.9.0-1_h8933c1f_netlib
libcblas               conda-forge/win-64::libcblas-3.9.0-5_hd5c7e75_netlib
libflang               conda-forge/win-64::libflang-11.0.1-h0e60522_20210131
liblapack              conda-forge/win-64::liblapack-3.9.0-5_hd5c7e75_netlib
llvm-openmp            conda-forge/win-64::llvm-openmp-11.0.1-h2d74725_0
m2w64-gcc-libgfortran conda-forge/win-64::m2w64-gcc-libgfortran-5.3.0-6
m2w64-gcc-libs         conda-forge/win-64::m2w64-gcc-libs-5.3.0-7
m2w64-gcc-libs-core   conda-forge/win-64::m2w64-gcc-libs-core-5.3.0-7
m2w64-gmp              conda-forge/win-64::m2w64-gmp-6.1.0-2
m2w64-libwinpthread   conda-forge/win-64::m2w64-libwinpthread-git-5.0.0.4634.697f757-2
msys2-conda-epoch     conda-forge/win-64::msys2-conda-epoch-20160418-1
python_abi             conda-forge/win-64::python_abi-3.9-2_cp39
slycot                 conda-forge/win-64::slycot-0.5.3-py39h73a3b6d_0
ucrt                   conda-forge/win-64::ucrt-10.0.22621.0-h57928b3_0

The following packages will be REMOVED:

numpy-base-1.23.5-py39h4da318b_0

The following packages will be UPDATED:

ca-certificates        pkgs/main::ca-certificates-2022.10.11~ --> conda-forge::ca-certificates-2022.12.7-h5b45459_0
numpy                  pkgs/main::numpy-1.23.5-py39h3b20f71_0 --> conda-forge::numpy-1.24.1-py39hbccbf6a_0
openssl                pkgs/main::openssl-1.1.1s-h2bbff1b_0 --> conda-forge::openssl-1.1.1s-hcfcfb64_1
vs2015_runtime         pkgs/main::vs2015_runtime-14.27.29016~ --> conda-forge::vs2015_runtime-14.32.31332-h1d6e394_9

The following packages will be SUPERSEDED by a higher-priority channel:

certifi                pkgs/main/win-64::certifi-2022.12.7-p~ --> conda-forge/noarch::certifi-2022.12.7-pyhd8ed1ab_0

Proceed ([y]/n)? y

Downloading and Extracting Packages

Preparing transaction: done
Verifying transaction: done
Executing transaction: done

(enel441) C:\Users\arne.dankers2>

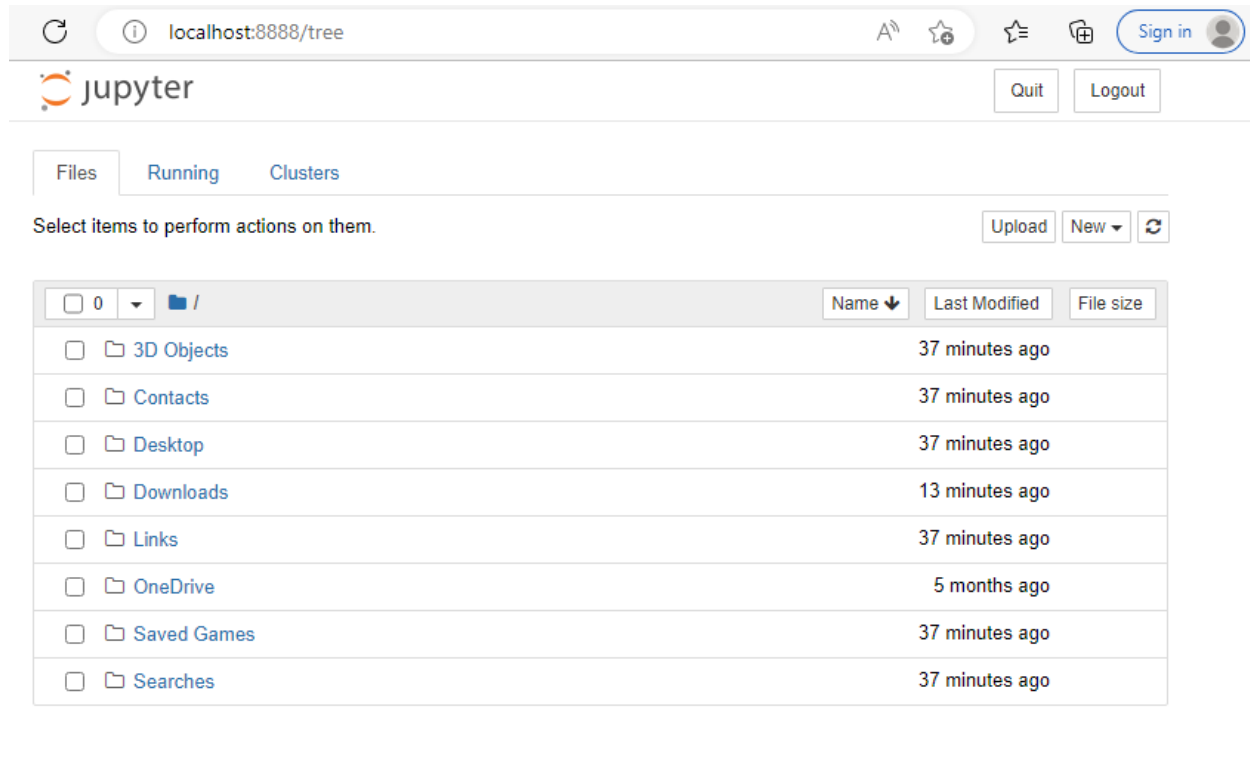
```

Now we have completed installing the python packages that we will need. In the following steps we will see how to launch a Jupyter Notebook. Make sure the enel441 environment is active.

At the miniconda prompt type:

```
>>jupyter notebook
```

This will open a window in your browser. Or it might open a new browser. The browser page will look like:



localhost:8888/tree

jupyter

Quit Logout

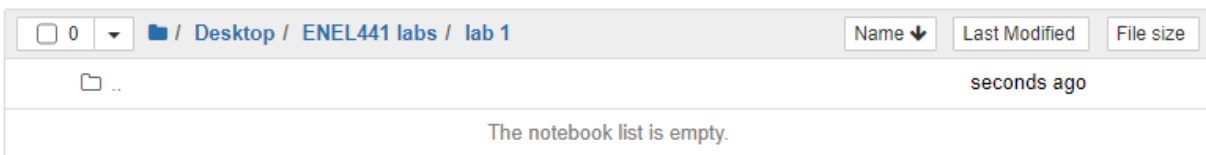
Files Running Clusters

Select items to perform actions on them.

Upload New

|                          | Name        | Last Modified  | File size |
|--------------------------|-------------|----------------|-----------|
| <input type="checkbox"/> | /           |                |           |
| <input type="checkbox"/> | 3D Objects  | 37 minutes ago |           |
| <input type="checkbox"/> | Contacts    | 37 minutes ago |           |
| <input type="checkbox"/> | Desktop     | 37 minutes ago |           |
| <input type="checkbox"/> | Downloads   | 13 minutes ago |           |
| <input type="checkbox"/> | Links       | 37 minutes ago |           |
| <input type="checkbox"/> | OneDrive    | 5 months ago   |           |
| <input type="checkbox"/> | Saved Games | 37 minutes ago |           |
| <input type="checkbox"/> | Searches    | 37 minutes ago |           |

Make a folder. I made one on the desktop called ENEL441, then I made another folder in ENEL441 called Lab1. Use the mouse to click to the folder you wish to store your notebook:



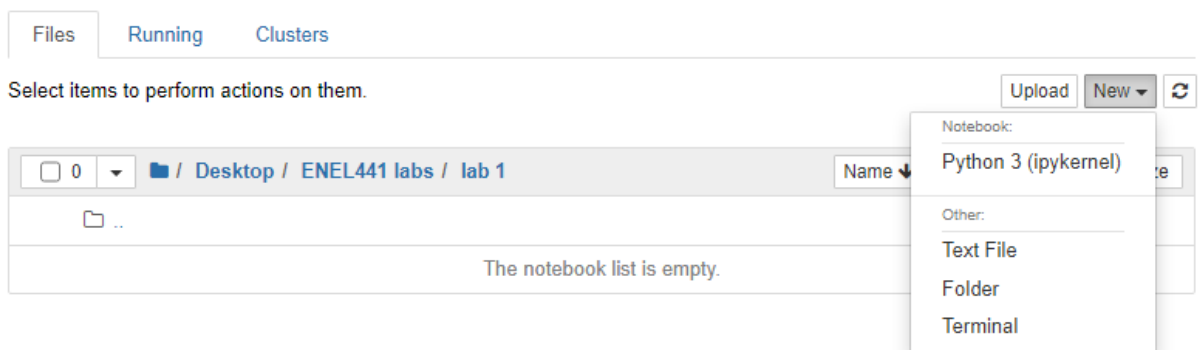
0 / Desktop / ENEL441 labs / lab 1

Name Last Modified File size

|                          |    |             |  |
|--------------------------|----|-------------|--|
| <input type="checkbox"/> | .. | seconds ago |  |
|--------------------------|----|-------------|--|

The notebook list is empty.

Then click on the 'New' button:



Files Running Clusters

Select items to perform actions on them.

Upload New

|                          | Name                             | Last Modified | File size |
|--------------------------|----------------------------------|---------------|-----------|
| <input type="checkbox"/> | / Desktop / ENEL441 labs / lab 1 |               |           |
| <input type="checkbox"/> | ..                               |               |           |

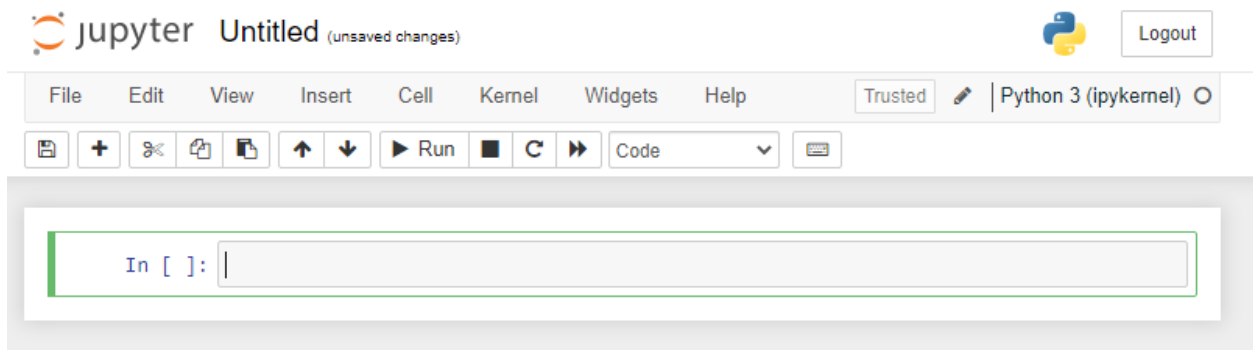
The notebook list is empty.

Notebook:  
Python 3 (ipykernel)

Other:  
Text File  
Folder  
Terminal

And select Python 3 (ipykernel)

This should start a new notebook. :



Click on 'File' then 'save as' to rename and save the file.

You are now ready to run python code in the notebook!

## Downloading the lecture notebooks from Git

You will need to download the notebooks for the lectures. To download the files from Github using the anaconda prompt follow these instructions:

At the Anaconda prompt, activate the enel441 virtual environment (if it is not already active):

```
>> conda activate enel441
```

Then change the directory to where you want the files to be downloaded. You can make a new directory using the mkdir command. Then use the cd command to go to that directory:

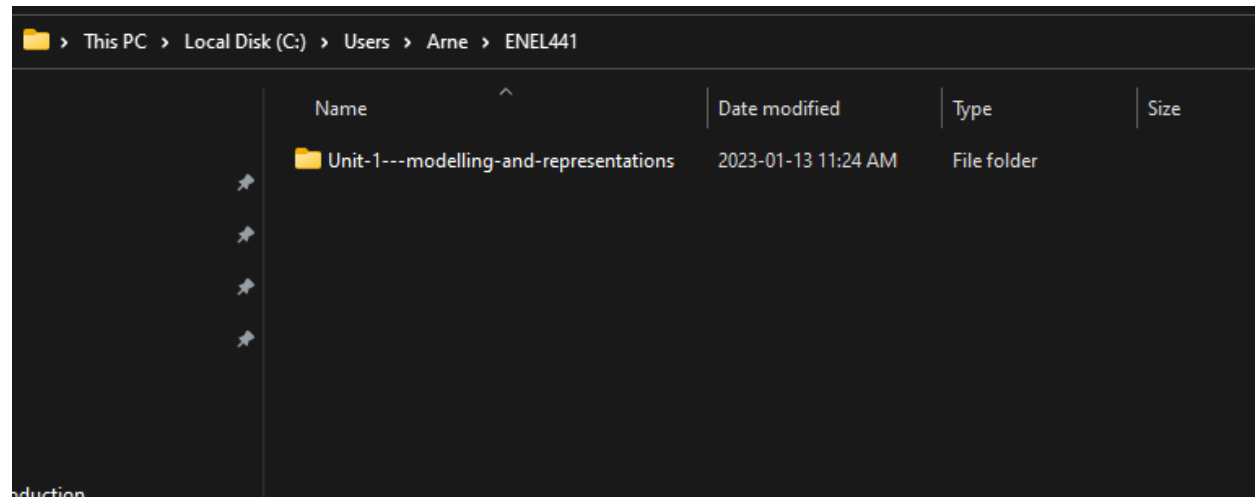
```
(enel441) C:\Users\Arne>mkdir ENEL441
(enel441) C:\Users\Arne>cd ENEL441
(enel441) C:\Users\Arne\ENEL441>
```

Now we will clone the git repository. Cloning a repository essentially means downloading the files from the repository to the directory you specified:

```
>>git clone https://github.com/arneUofC/Unit-1---modelling-and-representations
```

```
(enel441) C:\Users\Arne\ENEL441>git clone https://github.com/arneUofC/Unit-1---modelling-and-representations
Cloning into 'Unit-1---modelling-and-representations'...
remote: Enumerating objects: 82, done.
remote: Counting objects: 100% (82/82), done.
remote: Compressing objects: 100% (66/66), done.
remote: Total 82 (delta 31), reused 66 (delta 15), pack-reused 0
Receiving objects: 100% (82/82), 2.24 MiB | 2.76 MiB/s, done.
Resolving deltas: 100% (31/31), done.
(enel441) C:\Users\Arne\ENEL441>
```

Now in my File Explorer, I see that a new folder has been created in ENEL441:



In this folder you will find the contents of the repository, including the notebook files that we will use during lectures. Notebook files have extensions `.ipynb`.

You can now launch Jupyter Notebooks (using the instructions above) and open any of the `.ipynb` files.

## Using Google Colab

Google Colab allows you to connect to a GitHub repository and run the notebooks in the repo. Colab is hosted in the cloud by Google. The advantage of running notebooks in this way is that you do not need to install or download any software.

Go to the Google Colab website: <https://colab.research.google.com/>

You will see a welcome screen:

The screenshot shows the Google Colaboratory interface. At the top, the URL is `colab.research.google.com`. The page has a dark theme. On the left is a sidebar with a 'Table of contents' and a search bar. The main content area is titled 'Welcome to Colab!' and includes a video thumbnail titled '3 Cool Google Colab Features'. Below the video, there is a section 'What is Colab?' with a bulleted list of features.

colab.research.google.com

Welcome To Colaboratory

File Edit View Insert Runtime Tools Help

Table of contents

- Getting started
- Data science
- Machine learning
- More Resources
- Featured examples
- Section

Welcome to Colab!

If you're already familiar with Colab, check out this video to learn about interactive tables, the executed code history view, and the command palette.

3 Cool Google Colab Features

What is Colab?

Colab, or "Colaboratory", allows you to write and execute Python in your browser, with

- Zero configuration required
- Access to GPUs free of charge
- Easy sharing

Whether you're a **student**, a **data scientist** or an **AI researcher**, Colab can make your work easier. Watch [Introduction to Colab](#) to learn more, or just get started below!

Click on File -> open notebook. A popup will appear:

The screenshot shows a popup window with tabs for 'Examples', 'Recent', 'Google Drive', 'GitHub', and 'Upload'. The 'Recent' tab is selected, showing a list of notebooks. The list has columns for 'Title', 'Last opened', 'First opened', and an icon column. The notebooks listed are 'Welcome To Colaboratory', 'lecture\_1\_test\_binder.ipynb', 'Untitled4.ipynb', 'Unit 1 - Modelling and Simulation in Time Domain.ipynb', and 'Untitled3.ipynb'.

Examples Recent Google Drive GitHub Upload

Filter notebooks

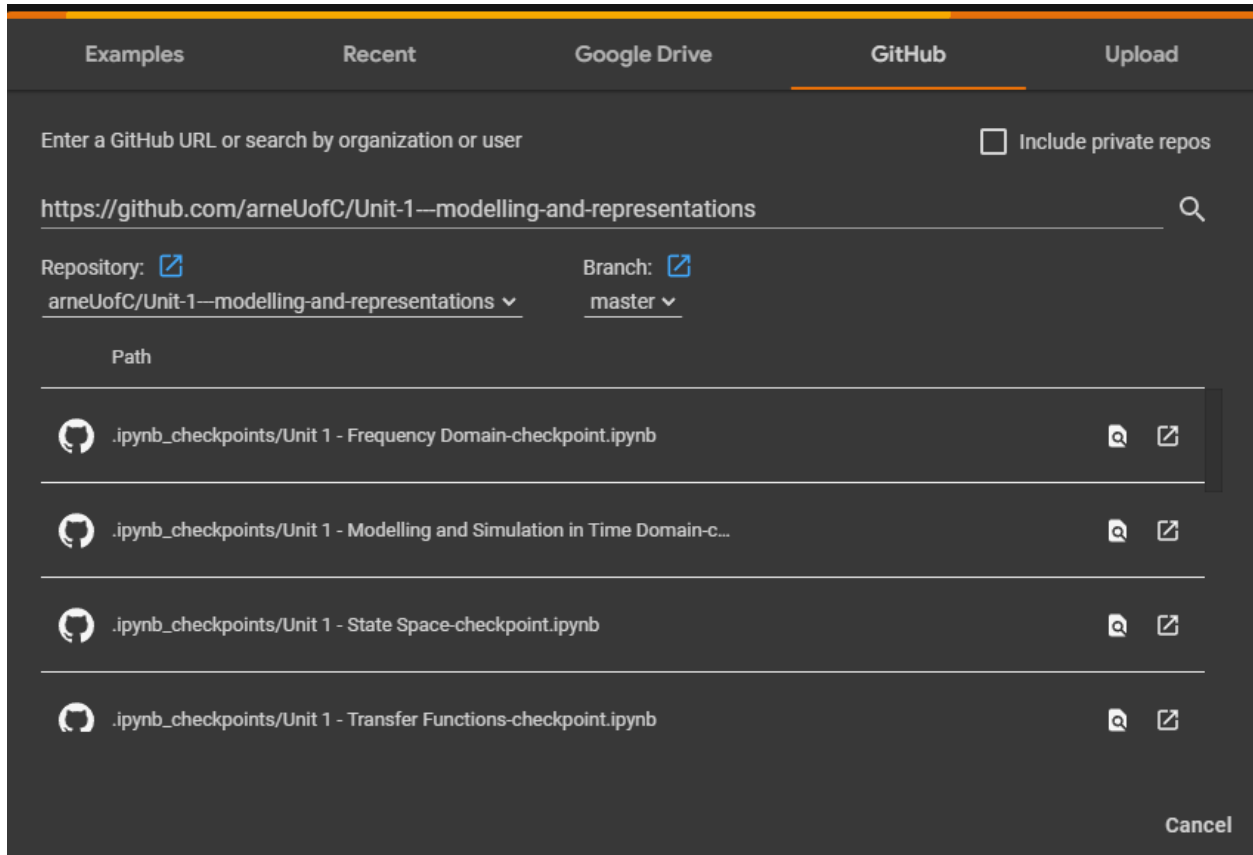
| Title                                                  | Last opened | First opened |  |
|--------------------------------------------------------|-------------|--------------|--|
| Welcome To Colaboratory                                | 10:35 AM    | Jan 23, 2020 |  |
| lecture_1_test_binder.ipynb                            | 10:27 AM    | January 10   |  |
| Untitled4.ipynb                                        | 10:16 AM    | 10:16 AM     |  |
| Unit 1 - Modelling and Simulation in Time Domain.ipynb | 10:09 AM    | 10:09 AM     |  |
| Untitled3.ipynb                                        | May 5, 2020 | May 5, 2020  |  |

Cancel

Click on the GitHub tab.

In the GitHub URL enter <https://github.com/arneUofC/Unit-1---modelling-and-representations>

You should see all the files in the repo:



Find the file 'Unit 1 – Modelling and Simulation in Time Domain.ipynb'

Click on it to open it.

You should see:

← → ↺

colab.research.google.com/github/arneUofC/Unit-1---modelling-and-representations/blob/master/Unit%201%20-%20Modelling%20and%20Simulation%20in%20Time%20Domain.ipynb

co

Unit 1 - Modelling and Simulation in Time Domain.ipynb

File Edit View Insert Runtime Tools Help

+ Code + Text Copy to Drive

Unit 1 - Modelling and Representations of Systems

Outline - Unit 1 - Modelling and Representations of Systems

- Modelling of electrical and mechanical systems
  - Electric circuits with resistors, capacitors, inductors, op-amps
  - Translational mechanical systems
  - Rotational mechanical systems
- Simulation of a system
  - Response due to initial conditions
  - Step response
  - Response to any input
- Representations
  - Differential equation
  - Transfer function
  - State-Space equations
  - Frequency domain
- Characteristics of a system
  - Settling Time
  - Maximum overshoot
  - Rise time
  - Stability
- Key Concept